Basics of The Koopman Operator

Institute of Control Systems

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PHD Workshop

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Introduction

- A reformulation of dynamical systems theory in terms of evolution of observables.
- Koopman and von Neumann used it to study measure-preserving systems.





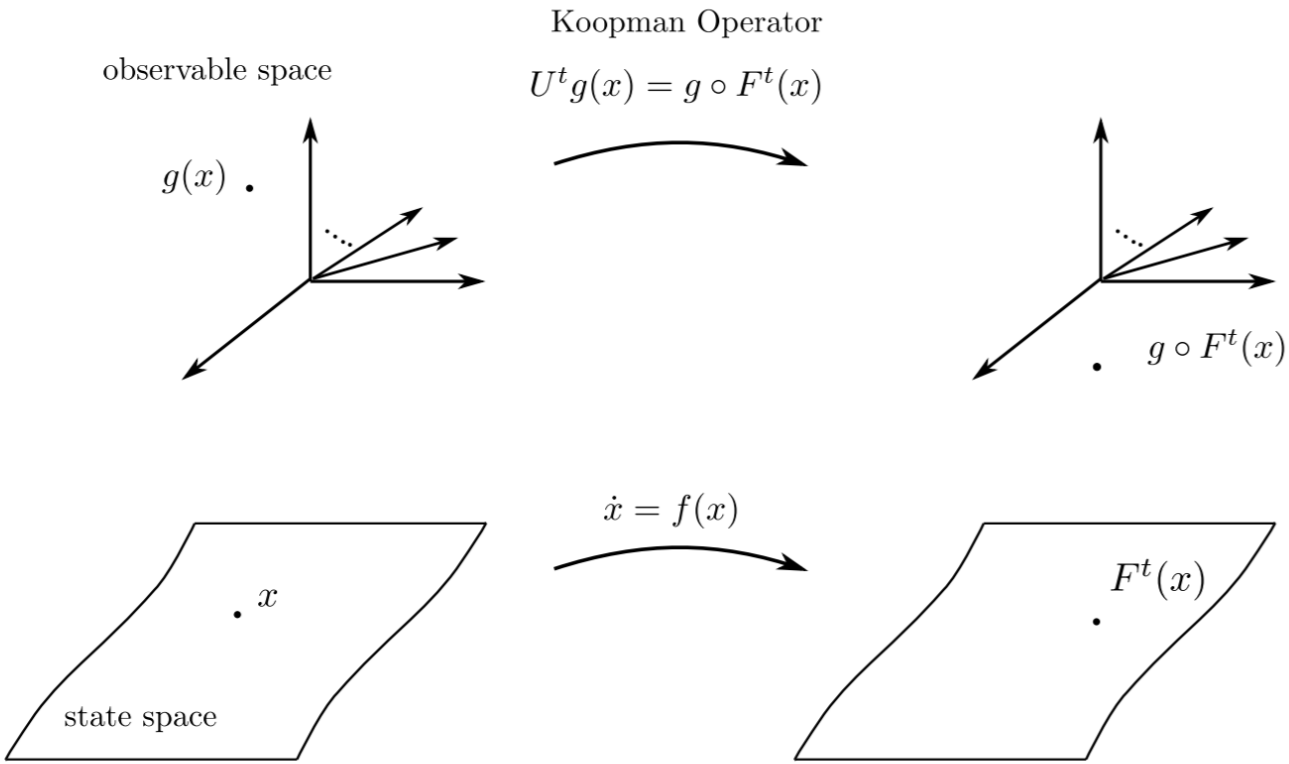


Figure: State-space to Infinite dimensional observable space

Continuous-time framework

$$\dot{\mathbf{x}}(t) = \mathbf{F}(\mathbf{x}(t)), \quad \mathbf{x}(0) = \mathbf{x}_0 \tag{1}$$

$$\mathbf{x}(t) = \mathbf{S}^{\mathbf{t}}(\mathbf{x_0}) \tag{2}$$

where $\mathbf{x} \in \mathbb{X} \subset \mathbb{R}^n$ and $\mathbf{S}^{\mathbf{t}}$ is the flow rate. Obserables $f \in \mathscr{H}$ is a map $f : \mathbb{X} \to \mathbb{R}$. The Koopman operator $U^t : \mathscr{H} \to \mathscr{H}$ is then defined by the relation

$$f(t, \mathbf{x_0}) = U^t f(\mathbf{x_0}) = f(\mathbf{S}^{\mathbf{t}}(\mathbf{x_0})) \tag{3}$$

$$\frac{\partial f(t,\mathbf{x})}{\partial t} = \mathbf{F}(\mathbf{x}).\Delta f(t,\mathbf{x}) = L_U f(\mathbf{x}) \tag{4}$$

L_U is the infinitesimal generator of the semigroup of Koopman operators $U^t{}_{t\geq 0} \implies U^t = e^{L_U t}$.

$$U^t \phi(\mathbf{x_0}) = \phi(\mathbf{S}^{\mathbf{t}}(\mathbf{x_0})) = e^{\lambda t}\phi(\mathbf{x_0}) \tag{5}$$

$$\frac{d\phi(t,\mathbf{x})}{dt} = \mathbf{F}(\mathbf{x}).\Delta\phi(\mathbf{x}) = \lambda\phi(\mathbf{x}) \tag{6}$$

Examples

$$\dot{\mathbf{x}}_1 = \mu \mathbf{x}_1$$

$$\dot{\mathbf{x}}_2 = \lambda(\mathbf{x}_2 - \mathbf{x}_1^2)$$

obervables: $f_1 = x_1 \quad f_2 = x_2 \quad f_3 = x_1^2 \implies$

$$\begin{bmatrix} \dot{f}_1 \\ \dot{f}_2 \\ \dot{f}_3 \end{bmatrix} = \begin{bmatrix} \mu & 0 & 0 \\ 0 & \lambda & -\lambda \\ 0 & 0 & 2\mu \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}$$

$$\dot{\mathbf{x}} = \beta \mathbf{x}(\mathbf{1} - \mathbf{x})$$

observable: $f_1 = x \quad f_2 = x^2 \implies$

$$\begin{bmatrix} \dot{f}_1 \\ \dot{f}_2 \end{bmatrix} = \begin{bmatrix} \beta & -\beta \\ ? & ? \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}$$

observable: $f_1 = x \quad f_2 = x^2 \quad f_{=} x^3 \implies$

$$\begin{bmatrix} \dot{f}_1 \\ \dot{f}_2 \\ \dot{f}_3 \end{bmatrix} = \begin{bmatrix} \beta & -\beta & 0 \\ 0 & \beta^2 & -2\beta^2 \\ ? & ? & ? \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}$$

Problems

Selection of observable to get invariant Koopman operator?

- Radial basis functions
- Logistic Functions
- Eigen functions
- polynomials

How to learn theses observable from data?

- Deep learning
- Hankel–DMD

Eigen functions

$$\dot{x} = f(x) = Ax + v(x)$$

1-Construction of Eigenfunctions for Nonlinear Dynamics The linearization of the dynamics around the origin is

$$\dot{\mathbf{y}} = \mathbf{D}f(0)\mathbf{y} = \hat{A}\mathbf{y}, \quad \mathbf{y} \in \mathscr{Y}$$

Proposition 1. *Let $\hat{A}_1$ denote the linearization (5) of the nonlinear system (1) with $\mathcal{Y}$ scaled into the unit hypercube, $\mathcal{Y}_1 \subset \mathcal{Q}_1$, and let $\{\mathbf{v}_1, \ldots, \mathbf{v}_d\}$ be a basis of the eigenvectors of $\hat{A}_1$ corresponding to nonzero eigenvalues $\{\lambda_1, \ldots, \lambda_d\}$.*

Let $\{\mathbf{w}_1, \ldots, \mathbf{w}_d\}$ be the adjoint basis to $\{\mathbf{v}_1, \ldots, \mathbf{v}_d\}$ such that $\langle \mathbf{v}_j, \mathbf{w}_k \rangle = \delta_{jk}$ and $\mathbf{w}_j$ is an eigenvector of $\hat{A}_1^*$ at eigenvalue $\bar{\lambda}_j$. Then, the linear functional

$$\psi_j(\mathbf{y}) = \langle \mathbf{y}, \mathbf{w}_j \rangle \tag{6}$$

is a nonzero eigenfunction of $U_{\hat{A}_1}$, the Koopman operator associated to $\hat{A}_1$. Furthermore, for any tuple $(m_1, \ldots, m_d) \in \mathbb{N}_0^d$

$$\left(\prod_{j=1}^{d} e^{m_j \lambda_j}, \prod_{j=1}^{d} \psi_j^{m_j} \right) \tag{7}$$

is an eigenpair of the Koopman operator $U_{\hat{A}_1}$.

Eigen functions

Proposition 2. *Assume that the nonlinear system (1) is topologically conjugate to the linearized system (5) via the diffeomorphism $h : \mathcal{X} \to \mathcal{Y}$. Let $B \in \mathcal{X}$ be a simply connected, bounded, positively invariant open set in $\mathcal{X}$ such that $h(B) \subset Q_r \subset \mathcal{Y}$, where Q_r is a cube in $\mathcal{Y}$. Scaling Q_r to the unit cube Q_1 via the smooth diffeomorphism $g : Q_r \to Q_1$ gives $(g \circ h)(B) \subset Q_1$. Then, if ψ is an eigenfunction for $U_{\hat{A}_1}$ at e^{λ}, then $\psi \circ g \circ h$ is an eigenfunction for U_f at eigenvalue e^{λ}, where U_f is the Koopman operator associated with the nonlinear dynamics (1).*

Example

$$\begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \\ \dot{y}_3 \end{bmatrix} = \underbrace{\begin{bmatrix} \mu & 0 & 0 \\ 0 & \lambda & -\lambda \\ 0 & 0 & 2\mu \end{bmatrix}}_{K} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\psi_1(\mathbf{y}) = \langle \mathbf{y}, \mathbf{w}_1 \rangle = y_1 = x_1$$
$$\psi_2(\mathbf{y}) = \langle \mathbf{y}, \mathbf{w}_2 \rangle = y_3 = x_1^2$$
$$\psi_3(\mathbf{y}) = \langle \mathbf{y}, \mathbf{w}_3 \rangle = y_2 + \frac{\lambda}{\lambda - 2\mu} y_3 = x_2 + \frac{\lambda}{\lambda - 2\mu} x_1^2$$

Example

$$\begin{bmatrix}\dot{\hat{x}}_1\\ \dot{\hat{x}}_2\end{bmatrix} = \underbrace{\begin{bmatrix}\mu & 0\\ 0 & \lambda\end{bmatrix}}_{A}\begin{bmatrix}\hat{x}_1\\ \hat{x}_2\end{bmatrix}$$

$$c(\mathbf{x}) = \begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ \frac{\lambda}{\lambda-2\mu}x_1^2\end{bmatrix}$$

$$\phi_1(\mathbf{x}) = \hat{\psi}_1(c(\mathbf{x})) = x_1$$

$$\phi_2(\mathbf{x}) = \hat{\psi}_2(c(\mathbf{x})) = x_2 + \frac{\lambda}{\lambda-2\mu}x_1^2$$

$$\phi_i(\mathbf{x}) = \hat{\psi}_i(c(\mathbf{x})) = x_1^{m_i^{(1)}}\left(x_2 + \frac{\lambda}{\lambda-2\mu}x_1^2\right)^{m_i^{(2)}}, \; i = 3, \ldots$$

Frame Title

Thanks for your attention!